ILLUMINATING A HIGH DIMENSIONAL BIOLOGICAL CYCLE

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Abstract.

1. Problem Statement and Motivation

The advent of genome-wide transcriptomics unveiled hitherto untapped reservoirs of vast biological data. These data - namely RNA transcripts produced within the cellular environment - provide direct insight into the mechanisms of cellular life and function. Analyzing and understanding trends and patterns within these extensive transcriptomic datasets thus becomes an important challenge in obtaining a better understanding of cellular life.

The scale and dimensionality of transciptome data are extreme. At any given moment, a cell may be expressing several thousand genes, and the number of transcripts in a cell's transcriptome is likely in the hundreds of thousands. Clever and robust data analysis techniques must be effectively deployed in order to coax out biologically realistic trends from such high-dimensional data and noise. In this paper, we attempt to apply several data dimension reduction and analysis techniques on a buddying yeast *Saccharomyces Cerecisiae* transcriptome dataset. These data, which consist of a total RNA transcriptome of a budding yeast cell prepared at 25 minute intervals, provide insight into how the transcriptome of yeast changes temporally. Specifically, our analysis attempts to use various techniques to determine the period of respiratory bursts in a synchronized yeast population's metabolic cycle and to determine which genes are expressed during the various phases of the metabolic cycle. Our investigation parses through a comlex transcriptomic dataset to provide insights into the specific cellular mechanisms of yeast metabolism, but also highlights advantages and drawbacks of various data analysis techniques on a high-dimensional, periodic dataset.

2. Data

Our dataset consists of a sequence of yeast microarray assays. Each microarray was designed to measure the abundance of the same set of 6,400 yeast gene transcripts along with other RNA transcripts of interest, bringing

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the total to 9725 transcriptomic features per assay. These assays were performed separately on 36 samples taken at regular 25-minute intervals from a metabolically oscillating yeast population. Our dataset is publicly available at the Gene Expression Omnibus here and comes from a peer-reviewed publication in *Science* here.

Even though microarrays are designed to be identical assays, microarray data is still subject to variability due to variations in staining and other experimental parameters. In the supplementary information, the authors of the dataset describe median-normalizing the raw intensity values for each gene (corresponding roughly with transcript concentrations) to reduce noise-based differences in total expression levels, which could have arisen due to differences in staining, etc. Although this preprocessing step was a standard practice at the time of this publication, it does rely on the assumption that the true median expression intensities for every chip were identical. This assumption means that our inference capabilities are limited - if the yeast truly produced more transcripts in total for some time points, resulting in a median, this difference would be masked by the median normalization. As a result, our data represents relativized rather than absolute expression levels, and we are unable to prove that the yeast produced absolutely more or less of a transcript from one time point to another, only whether each gene was more or less expressed at a timestep relative to the median transcript intensity at that timestep.

The 9725 time series produced by these normalized microarray results represent enormous ranges of mean and variance, with some transcripts expressed several orders of magnitude more than others. To attempt to make our median-normalized expression time series more comparable, we took an arcsinh transformation of all intensity values. This is similar to log-transforming data but with better handling of zeros, which are common in transcriptomics. Although improvements to this process such as transforming with a learned affine function prior to arcsinh could be made (see here), visual inspection of the data confirmed that the data were more comparable and we decided to proceed.

The arcsinh data was used for our analyses that require comparing relative sizes of expression between different genes, but for our analyses that cluster genes, it also made sense to standardize the expression values of each gene by the mean and standard deviation of that gene.

3. Methods

The paper we are building upon emphasized the periodic nature of many of the genes in the dataset. As such, our analysis included an independent verification of a suitable period length, a process to filter genes by periodicity, and additional analyses sought to better understand the periodic modules in our data.

3.1. **Period Length Identification.** We fit Gaussian Mixture Models (GMMs) to the full set of arcsinh-transformed gene expressions to identify clusters of co-expressed genes and to recover the underlying period of the yeast metabolic cycle.

Approach 1: High-variance feature selection. A naive GMM fit on all 9,275 genes as features suffers from the curse of dimensionality ($n \ll d$), producing overconfident, nearly deterministic cluster assignments. To try to mitigate this, we discarded all but the n highest variance genes (rows) and fit a 4-component GMM on all 36 time points. The intuition is that high-variance time points are the most discriminative for separating phases of the metabolic cycle. We swept n from 200 to 1100 in steps of 100, observing that cluster assignments were stable and reflected a roughly cyclic sequence consistent with a 300 s period, or three total periods in the experiment.

However, even this large reduction did not produce real soft-clustering, as the assignments were still essentially deterministic. We decided to reduce to even less than 200 dimensions.

Approach 2: PCA + GMM on time steps. As a more principled dimensionality reduction, we projected the 9,275-gene feature space down to 5 principal components and fit a GMM with tied covariance on the reduced representation. The data were split temporally — the first 24 time points for training and the remaining 12 for testing — to avoid leakage. We selected the number of components $k \in 2, \ldots, 7$ by evaluating held-out log-likelihood on the test set over 100 random restarts per k, finding $k = 4$ to be consistently well-supported. Predictions over the full 36-time-point sequence show a repeating cluster sequence, directly recovering the oscillation period.

Interestingly enough, even with an observation space of just 5 dimensions and a hidden state space of just 4, the assignments are still very high probability.

Additionally, we fit a GMM on the 200 highest variance genes and then projected to 2 principal components, and graphed the 36 time steps colored by their GMM assignment. There are 4 distinct clusters in 2 dimensions which are correctly discovered by the GMM in high dimensions.

Conclusions. The most important takeaway of the GMM clustering is the recovery of the period of the yeast's metabolic cycle. Every method tried consistently recovered a 12 time-step period, or 300 seconds. This knowledge will be used in further analysis methods, such as for classical time series analysis, and the splitting of data into basically periodic and non-periodic.

To do still. The selection of four hidden states is essentially arbitrary. While it seems to work well, the selection should be made more rigorous, perhaps via a grid search. We will also fit a GMMHMM for inference and prediction. The question of why soft-clustering still stands as well, and we will either successfully soft-cluster or explain why we cannot.

3.2. **Periodic Gene Identification.** Once an overall period length was identified, we used permutation testing on autocorrelation to identify which of the 9725 genes were oscillatory on that period. We computed the autocorrelation of the 36-timestep time series for each gene with delay equal to the optimal period length. We then randomized the order of the intensities in each time series 100 times and computed the autocorrelations of these time series. If the autocorrelation a true series was at the 95th percentile or greater relative to the randomized baseline, we accepted the gene as likely periodic and gave it a periodic label for downstream analyses. The other genes were rejected as non-periodic, since even significant numbers of randomized permutations of their time series had similar autocorrelations at the optimal period delay.

3.3. **Classical Time Series Analysis.** Once a suitable overall period and individual periodic genes were identified, we used ideas from classical time series analysis to investigate the periodic genes specifically in more depth. We obtained the classical time series decomposition of each individual gene via moving averages over the 24 central time steps (as moving average requires rejecting timesteps at the beginning and end), and then clustered the seasonal components identified across these 24 timesteps using GMMs optimized for cluster count by inspecting the AIC and BIC curves. We then inspected the individual clusters in more depth through visualization.

TODO: Semantic Evaluation of Clusters via AI model

TODO: How should we incorporate a train/test split?

3.4. **ARMA.**

3.5. **Other analyses?**

4. Results and Analysis

4.1. **Period Length Identification.** * a lot of this appears to be above in methods - we might move some of it down here*

... a period of 12 timesteps, corresponding to $25 \times 12 = 300$ minutes, was the clear choice in all cases.

4.2. **Periodic Gene Identification.** We performed our permutation testing to identify the genes with significant 12-timestep delay autocorrelations, in line with our conclusion that a 12 timestep period was most plausible. Via this analysis, 5243 of the genes - over half of all genes - were identified as periodic with the 12 timestep period. Dimensionality reduction via PCA and visualization showed a clear difference in the distributions of periodic and non-periodic genes.

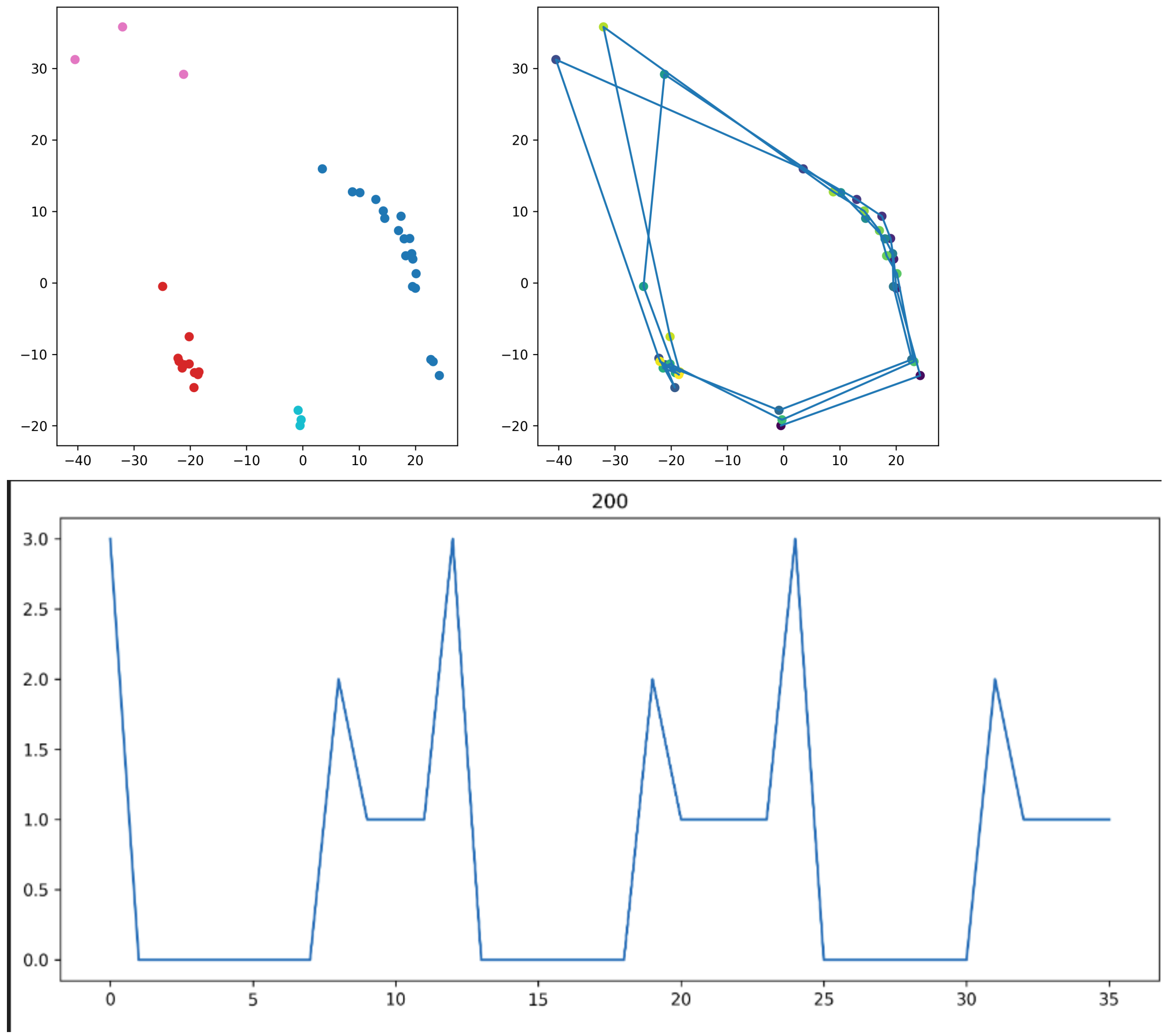

Figure 1. Temporal cycles in cell state. [TODO: Cleanup and label] Upper Left: The 200 genes with the highest variance in the arcsinh transformed data were extracted and clustered with a GMM with four components, then the data dimensionality was reduced via PCA. Upper Right: The same plot with a line connecting adjacent timesteps. Bottom: When visualized, a clearly periodic pattern was clear in cluster assignment, repeating every 12 timesteps.

4.3. **Classical Time Series Analysis.** After extracting the seasonal decomposition for each gene's time series using the mean- and variance-standardized data, we attempted to cluster these seasonal components using GMMs. We chose eight clusters, because this minimized the BIC and gave a local minima in the AIC.

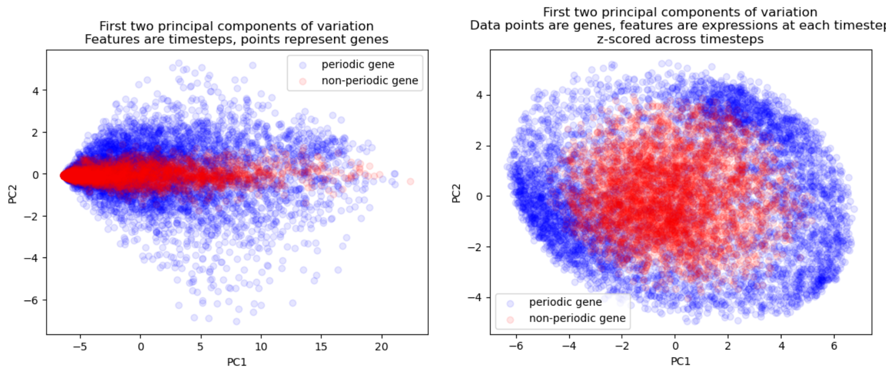

Figure 2: Distributional differences in the time series of periodic and non-periodic genes. [TODO: Cleanup and add better labels] After using autocorrelation and permutation testing to determine which genes were periodic, we used dimensionality reduction to visualize the genes. Left: Principal Component Analysis visualization of the arcsinh-transformed expression values for each gene across the 36 timesteps. It appears that the first principal component represents something similar to average arcsinh expression across timesteps, which makes sense because arcsinh does not remove the massive differences in average expression (just rescales them). Right: Same values but normalized via a StandardScaler applied to each time series individually. Now all genes have the same mean expression, and the distributions of periodic and non-periodic genes are significantly different in both principal components.

Visualization of the seasonal components of normalized expression by cluster revealed distinct temporal modules of genetic expression in all but one cluster, with peaks or troughs of different shapes and sizes at different phases of the cycle. Of these 7 clusters, visualization showed that two pairs appear to be especially similar to one another, suggesting that perhaps a more natural cluster number would be 5 or 6. The remaining cluster appeared to represent a catchall for noisy time series that did not have a clear choice in cluster.

4.4. **ARMA.**

4.5. **Other analyses?**

5. Ethical Implications and Conclusions

Interpretation and analysis of large genomic or transcriptomic data sets is becoming a canonical use case for the application of various data analysis techniques like the ones used here. Using a strategy of analyzing data using various techniques and at various levels of data dimension reduction can provide novel insights into the dataset being studied, and when several techniques converge to the same conclusion, our results obtain an additional

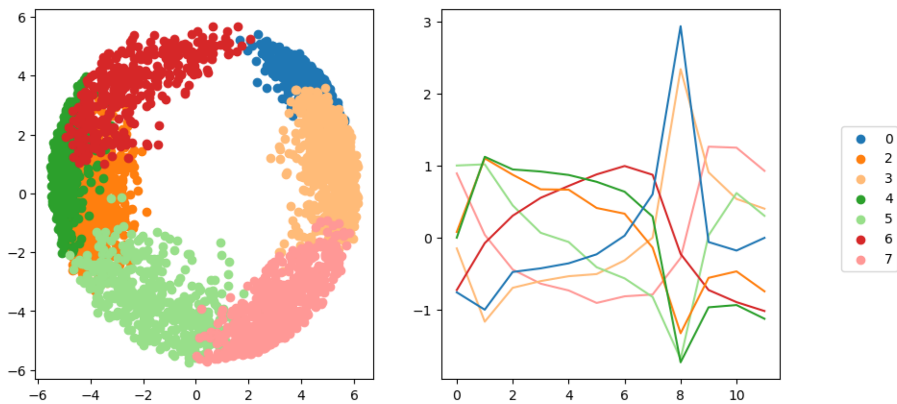

Figure 3: Temporal expression modules. [TODO: Cleanup + Label Graph] Left: UMAP of genes according to the seasonal component of their standardized periodic expressions, colored by GMM cluster. Right: The seven clearly consistent GMM clusters. The noisy cluster (1) was removed from the analysis.

robustness. In an era when the biological sciences are becoming increasingly under strain by a crisis of reproducibility and replication, the ability to produce consistent conclusions when reexamining and validating the analysis of a dataset is critical.

Obviously, mathematical evaluation of mathematical data does *not* address the crux of this crisis - replication, where an experiment is repeated to validate previous results. However, when replication is situationally challenging, developing robust strategies of reproducibility can strengthen the credibility of scientific researchers and strengthen their conclusions. Thus, deployment of various strategies to analyze complicated, high-dimensional data is important both to gain novel insights into experimental data, but also to validate previous conclusions about a dataset. Prudent, robust, and comprehnsive approaches to reproducibility lends additional credibility to experimental conclusions, and thus is a boon to the credibility of modern biological sciences as a whole.

6. Conclusion

Analyzing the data provided an independent confirmation of some of the Yeast paper's authors conclusions. For example, the period they identified from the data (300 minutes) for the yeast metabolic cycle oscillations via a maximum likelihood analysis was rediscovered by our independent use of GMMs to cluster the timesteps, and we like the authors identified slightly more than half of the dataset's genes as periodic with this period. This concordance reinforces the strength of those conclusions by the original authors about their data. We also were able to provide novel analyses and interpretations of the data that the authors did not reach. For example, our extraction of the seasonal component of the standardized time series and our use of GMMs to cluster it revealed 5-7 distinct periodic modules of activation, which improves upon the granularity of the authors' original analysis identifying only three clusters using k-means. This increased granularity could lead to improved biological insight. (Some conclusions about the success of any kind of predictions on the time series?)